

SYSTEM TUNING CHEAT-SHEET

English



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HushMonkey - SYSTEM TUNING CHEAT-SHEET

1. SETUP

- a. Position speakers and subwoofers correctly in the room.
- b. Place the measuring microphone at ear level in the coverage area (about 8–12 m from the stage).
- c. Ensure a healthy input signal in the software (green levels, no clipping).

2. TIMING (Top boxes)

- a. *Play Pink Noise through the top ONLY:*
- b. **Measure the top:** Open the Transfer Function.
- c. **Find Time:** Tap **Auto-Track** (or manually adjust until the IR peak is at 0ms the IR graph).
- d. **Lock Time Reference:** Press **Lock Delay**  .
- e. **Save Reference:** Take a **Capture** of the Peak's phase and frequency response.
- f. **Mute the top.**

3. PHASE ALIGNMENT (Subwoofer)

- a. *Turn on the subwoofer ONLY (DO NOT touch the Lock Delay!):*
- b. **Measure the Sub:** Observe the phase curve of the Sub in relation to your saved Top-trace.
- c. **Adjust Delay:** Use the +/- **0.1ms** buttons in the software until the Sub's phase line **sneaks parallel on top of the Top phase line** in the parts area (e.g. at).80 Hz
- d. **Transfer:** Read the detected delay and **enter it directly into your DSP/mixer on the Sub channel.**

4. VERIFICATION (Summary)

- a. *Turn on BOTH Top and Sub simultaneously (NEVER check with Auto-Track here!):*
- b. **Check the Magnitude:** You should see a nice sum increase (up to extra energy) in the overlap area. +6 dB
- c. **Check Phase:** The phase line should run smoothly and unbroken through the transition zone without sudden dips or punctures.

5. EQ & TONAL BALANCE (Spectrum & RTA)

- a. *Now the frequency measurement gives a truthful picture without phase noise:*
- b. **Check Room Resonances (Transfer/Spectrum):** Use narrow notch filters (EQ) to dampen standing waves/deep peaks.
- c. **Avoid "False Dives":** NEVER turn up the EQ in deep jumps if **the Coherence** is low (these are acoustic cancelations from the room, not an EQ issue!).
- d. **Tonal Balance:** Play familiar reference music and **check the Spectrum against your House Curve** (adjust the last broad EQ to make the sure the sound is comfortable).

6. SHOWTIME (Live Monitoring)

- a. During the concert, you switch to **Spectrum (RTA) & LAF-Waterfall (Spectrogram).**
- b. Keep an eye out for **vertical, continuous luminous lines** on the waterfall – these are feedback or space resonances hanging in the air. Mute the specific frequency of your EQ before it bursts!

With this note at hand, a full phase tuning and room correction will take you less than 5 minutes, and you are always sure to get the most out of your system!